Your results

Full Mock Exam

33.3% 40/120

40 out of 120 correct

Time taken 58:38

✓ Correct	✕ Wrong	– Not answered
40	40	40

Section	Score	
1 · Aptitude and Reasoning Skills	8/20	40%
2 · Language Skills	9/20	45%
3 · General Sciences	22/40	55.0%
4 · Mathematics	1/40	2.5%

Question breakdown

All (120) | Wrong (40) | Unanswered (40)

All sections

Question 1 · Aptitude and Reasoning Skills — ✕ Wrong

Pointing to a photograph of a boy, Ben said, "He is the son of the only son of my mother." How is Ben related to that boy?

A Brother

B Uncle ✕ Your answer

C Cousin

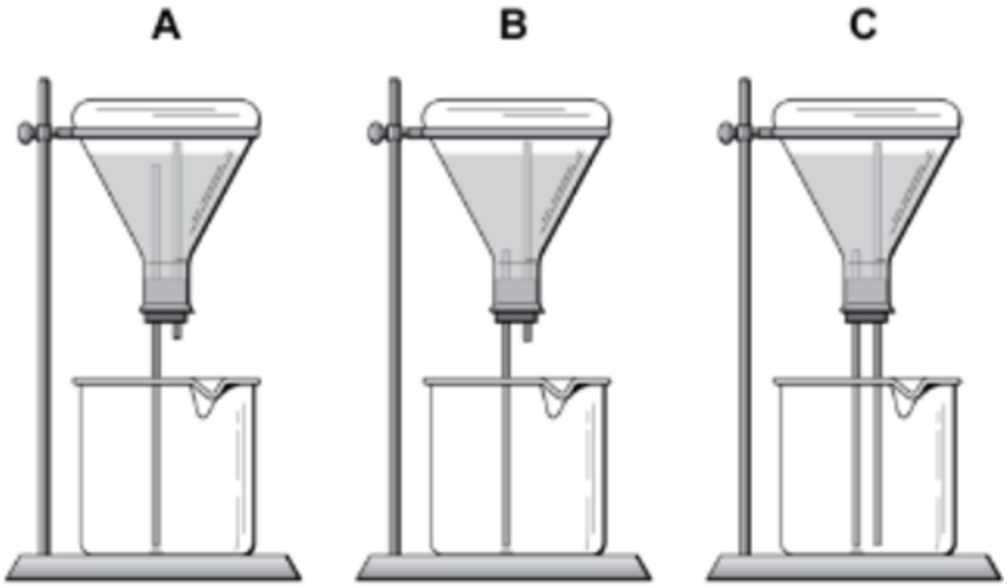
D Father ✓ Correct answer

Explanation

The "only son of my mother" refers to Ben himself, so "the son of [Ben]" is Ben's own son — making Ben the boy's father.

Question 2 · Aptitude and Reasoning Skills ✕ Wrong

If the top stoppers of all three systems are completely sealed airtight, which system will allow the most water to continuously run out into the beaker?



A System A

B System B

C System C ✓ Correct answer

D All systems will drain completely due to gravity ✕ Your answer

Explanation

With all systems sealed airtight, System C's configuration allows air to re-enter and equalize pressure as water drains, sustaining continuous flow while the others create a vacuum that stops flow.

Question 3 · Aptitude and Reasoning Skills — Not answered

Pointing to a photograph, a man said, "I have no brother or sister but that man's father is my father's son." Whose photograph was it?

A His father's

B His son's ✓ Correct answer

C His own

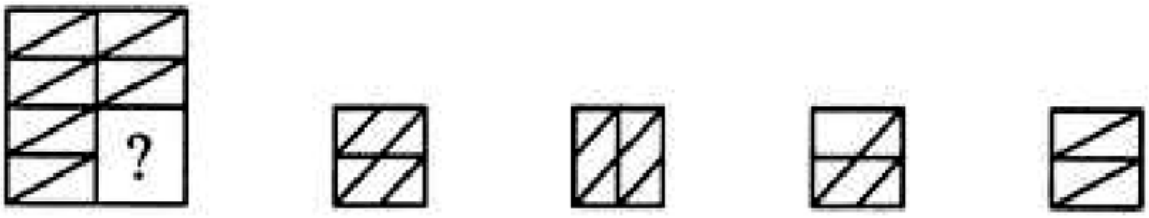
D His nephew's

Explanation

The man has no siblings, so "my father's son" refers to himself; therefore "that man's father is [the speaker]," meaning that man is his son.

Question 4 · Aptitude and Reasoning Skills ✓ Correct

Identify the figure that completes the pattern.



A 1

B 2

C 3

D 4 ✓ Your answer

Explanation

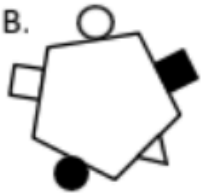
The diagonal bands continue with the same spacing and alignment into the missing quadrant. Figure 4 matches the continuation, corresponding to choice D.

Question 5 · Aptitude and Reasoning Skills ✓ Correct

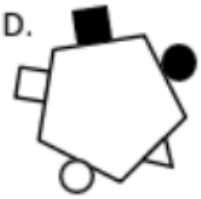
If you rotate the given figure, which of the following will be the result?



A. 

B. 

C. 

D. 

A Figure A

B Figure B

D Figure D

Explanation

A rotation preserves the clockwise order of the marks attached to the polygon. Figure C retains the positions and sequence of the black and white symbols without reflecting them.

Question 6 · Aptitude and Reasoning Skills ✗ Wrong

ISAAC NEWTON : LAW OF UNIVERSAL GRAVITATION :: ALEXANDER FLEMING : ?

A PENICILLIN ✓ Correct answer

B MONOGRAPH

C TELEPHONE ✕ Your answer

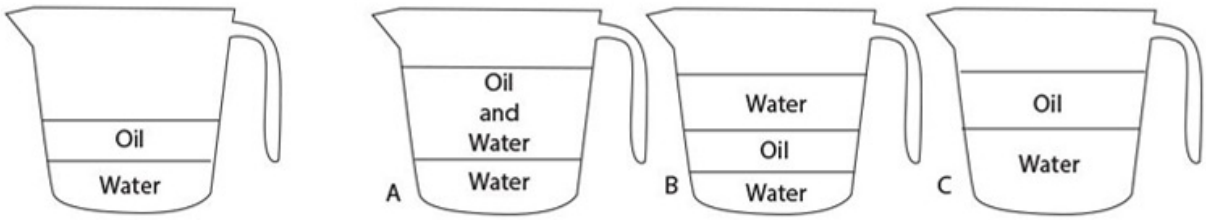
D SONOGRAM

Explanation

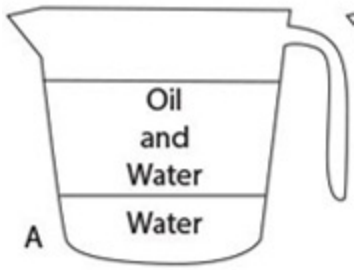
Isaac Newton formulated the law of universal gravitation, and Alexander Fleming discovered penicillin.

Question 7 · Aptitude and Reasoning Skills ✓ Correct

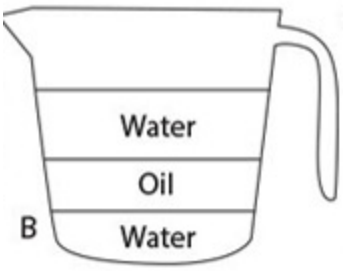
A glass beaker contains water and oil, if you add water, what would it look like?



A

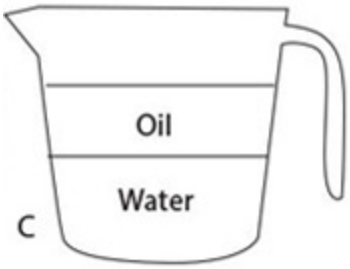
A 

B

B 

C

C ✓ Your answer



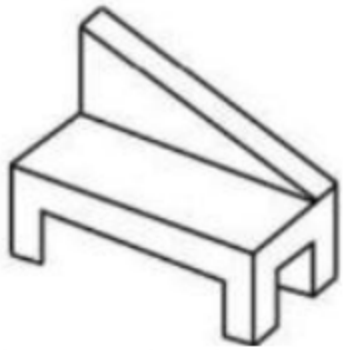
D None of the above.

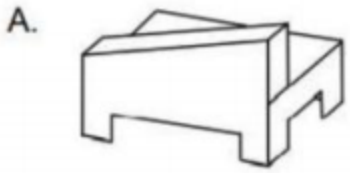
Explanation

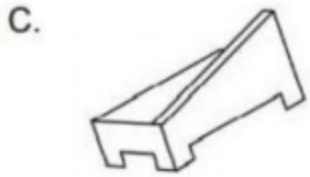
Water is denser than oil, so added water sinks and settles below the oil layer, keeping the two clearly separated with oil on top.

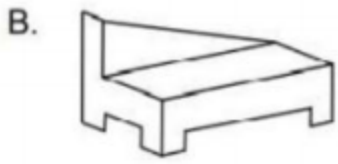
Question 8 · Aptitude and Reasoning Skills ✕ Wrong

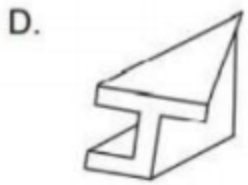
Which of the following is a rotation of the solid?











A Figure A ✕ Your answer

B Figure B ✓ Correct answer

C Figure C

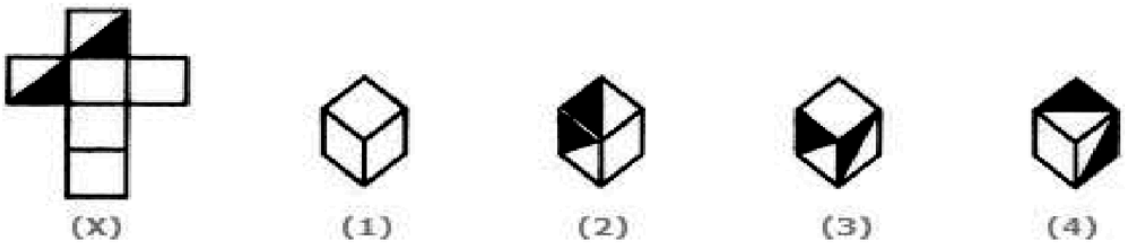
D Figure D

Explanation

Option B shows the same bench-like solid simply reoriented in space, without being mirrored, unlike the other options.

Question 9 · Aptitude and Reasoning Skills — ✕ Wrong

Choose the box that is similar to the box formed from the given sheet of paper (X).



A 1 and 4 — ✓ Correct answer

B 2 and 4 — ✕ Your answer

C 1 and 2

D 2 and 3

Explanation

Folding the cross-shaped net (X) along its lines produces boxes matching the triangle/shading placement shown in boxes 1 and 4, while 2 and 3 place the shaded sections in positions inconsistent with the net.

Question 10 · Aptitude and Reasoning Skills — ✕ Wrong

A three-dimensional object is shown. Which front side view is correct?

A Figure A — ✕ Your answer

B Figure B

C Figure C

D Figure D ✓ Correct answer

Explanation
The 3D object has a stepped, offset front face; option D is the only front view whose internal divisions correctly reflect that step pattern.

Question 11 · Aptitude and Reasoning Skills ✕ Wrong

MALARIA : MOSQUITO :: ? : ?

A POISON : DEATH

B CHOLERA : WATER ✓ Correct answer

C RAT : PLAGUE

D MEDICINE : DISEASE ✕ Your answer

Explanation
Malaria is transmitted via mosquitoes, and cholera is similarly transmitted through contaminated water — matching the disease: transmission-vector relationship.

Question 12 · Aptitude and Reasoning Skills ✓ Correct

Choose the word which is different from the rest.

A Chicken

B Swan

C Eagle

D Crocodile ✓ Your answer

Explanation
Chicken, swan, and eagle are all birds, while a crocodile is a reptile.

Question 13 · Aptitude and Reasoning Skills ✓ Correct

Choose the word which is different from the rest.

A Dagger

B Knife

C Hammer ✓ Your answer

D Sword

Explanation

Dagger, knife, and sword are all bladed/cutting weapons, while a hammer is a blunt tool.

Question 14 · Aptitude and Reasoning Skills ✕ Wrong

If a piece of paper containing two circular holes as shown is folded along the dashed lines, which of the following is the resulting figure?

A Figure A ✕ Your answer

B Figure B ✓ Correct answer

C Figure C

D Figure D

Explanation

The two holes begin in diagonally opposite quadrants. Folding the sheet along both dashed centerlines brings the holes into a single smaller square while preserving their diagonal arrangement. Therefore, option B, which shows two diagonally positioned holes, is correct.

Question 15 · Aptitude and Reasoning Skills ✓ Correct

In the diagram shown below, if block #1 is moved 10 feet to the right, how far upward is block #2 lifted?

A 3 feet

B 5 feet

C 10 feet ✓ Your answer

D 20 feet

Explanation

Fixed pulleys only redirect the cable's direction; they don't add mechanical advantage, so Block 2 rises the same distance the cable is pulled.

Question 16 · Aptitude and Reasoning Skills ✓ Correct

GALILEO GALILEI:TELESCOPE::BENJAMIN FRANKLIN:

A ELECTRICITY ✓ Your answer

B STEAM ENGINE

C TELEGRAPH WIRE

D MOTION PICTURE CAMERA

Explanation

Just as Galileo is associated with the telescope, Benjamin Franklin is famously associated with the discovery/study of electricity.

Question 17 · Aptitude and Reasoning Skills ✗ Wrong

Which of the following frames completes the pattern?

A Figure A

B Figure B ✗ Your answer

C Figure C ✓ Correct answer

D Figure D

Explanation

Following the diagonal line and shading progression across the grid, option C is the only frame that continues the pattern consistently into the missing cell.

Question 18 · Aptitude and Reasoning Skills — ✕ Wrong

Statements: No door is a dog. All dogs are cats. At least one dog exists. Conclusions: (1) No door is a cat. (2) No cat is a door. (3) Some cats are dogs. (4) All cats are dogs.

- A Only (1) and (4) — ✕ Your answer
- B Only (1) and (3)
- C Only (3) and (4)
- D Only (3) — ✓ Correct answer

Explanation

At least one dog exists, and every dog is a cat. Therefore, some cats are dogs, so only conclusion (3) follows. Nothing given rules out doors that are cats, and not every cat must be a dog.

Question 19 · Aptitude and Reasoning Skills — ✓ Correct

Choose the alternative which closely resembles the mirror image of the given combination.

- A 1 — ✓ Your answer
- B 2
- C 3
- D 4

Explanation

A vertical mirror reflection reverses the order and the orientation of the letters in EFFECTIVE. The correctly reflected word is shown in Figure 1, corresponding to choice A.

Question 20 · Aptitude and Reasoning Skills — ✕ Wrong

Choose a figure which would most closely resemble the unfolded form of Figure (Z).

A 1 ✕ Your answer

B 2

C 3 ✓ Correct answer

D 4

Explanation

Opening the folded corners reproduces the hole positions shown in Figure 3. The result must preserve the reflection of each puncture across the fold lines.

Question 21 · Language Skills ✕ Wrong

Rousseau advised, “Have a good bank account, a good cook and a good digestion.” What is being referred to?

A duty

B investment ✕ Your answer

C security ✓ Correct answer

D family

Explanation

The bank account represents financial stability, the cook represents reliable food and comfort, and good digestion represents physical well-being. Among the choices, “security” best brings these conditions together as a comfortable, well-provided life. Investment covers only the financial element, while duty and family are not the central idea of the quotation.

Question 22 · Language Skills ✕ Wrong

Identify the option containing an error. If there is no error, choose D.

Severe pollution, along with overfishing and habitat destruction, threaten marine biodiversity in the region.

A along with

B threaten ✓ Correct answer

C region

D No error ✕ Your answer

Explanation

The subject is singular “Severe pollution.” The added phrase beginning “along with” does not create a compound subject, so the verb should be “threatens.” Compare “Pollution, along with overfishing, threatens biodiversity” with the different construction “Pollution and overfishing threaten biodiversity.”

Question 23 · Language Skills ✓ Correct

Engineers tested the new bridge design many times before construction began. Which conclusion is most reasonable?

A Engineers wanted to ensure safety. ✓ Your answer

B Engineers dislike building bridges.

C The bridge was already broken.

D Construction workers refused to help.

Explanation

Testing a bridge design before construction helps engineers check whether it will perform safely. Thus, ensuring safety is the conclusion best supported by the sequence of events. The passage provides no evidence that the engineers dislike bridges, that the unbuilt bridge was already broken, or that workers refused to help.

Question 24 · Language Skills ✕ Wrong

Winning the DOST scholarship will help him burn the midnight oil to maintain his high grades. What does it mean?

A Waste valuable electricity

B Wake up early in the morning ✕ Your answer

C Study or work late into the night ✓ Correct answer

D Cook meals late at night

Explanation

“Burn the midnight oil” means study or work late into the night. In the scholarship context, it refers to spending late hours studying to maintain grades. The expression is figurative and does not mean wasting electricity, waking early, or preparing a late meal.

Question 25 · Language Skills ✕ Wrong

Choose the word closest in meaning to the capitalized word as used in the sentence.

Despite facing major setbacks during the experiment, the team remained RESILIENT.

A Defeated

B Indifferent

C Adaptable ✓ Correct answer

D Hesitant ✕ Your answer

Explanation

“Resilient” describes the ability to recover or keep functioning after difficulty. Among the choices, “adaptable” is closest because an adaptable team can adjust to setbacks. “Defeated” suggests giving up or being overcome, “indifferent” means unconcerned, and “hesitant” means uncertain about acting.

Question 26 · Language Skills ✕ Wrong

Choose the best revision. Consider correctness, clarity, coherence, and conciseness.

Our community project should succeed. The volunteers are inexperienced. The coordinator needs to provide clearer instructions.

A Although the volunteers are inexperienced, our community project should succeed with the coordinator clearer instructions. ✕ Your answer

B The volunteers are inexperienced; still, our community project could succeed with the coordinator giving clearer instructions.

C To ensure the success of our community project, the coordinator needs to provide clearer instructions to the volunteers. ✓ Correct answer

D The coordinator needs to provide clearer instructions to the volunteers, so our community project could succeed.

Explanation

C is concise and directly connects clearer instructions with project success, which explains the source's preference. However, it omits the fact that the volunteers are inexperienced. B retains that detail and is also grammatically acceptable, though less concise. The question needs a clearer priority between preserving every idea and shortening the statement to ensure a unique best answer.

Question 27 · Language Skills ✕ Wrong

Identify the option containing an error. Use the formal scientific convention in which “data” is plural. If there is no error, choose D.

The data collected by the field researchers suggests that environmental factors are affecting crop yields significantly.

A suggests ✓ Correct answer

B are

C significantly

D No error ✕ Your answer

Explanation

Under the specified formal scientific convention, “data” is plural, so the verb is “suggest.” The phrase “collected by the field researchers” modifies “data” and does not change its number. “Factors are” already has correct plural agreement. Singular “data suggests” occurs in other usage conventions, so the stated convention is essential to this item.

Question 28 · Language Skills ✕ Wrong

Choose the best revision. Consider correctness, clarity, coherence, and conciseness.

The Filipino Scientist Fe del Mundo invented medical devices for children, and she became known throughout the world.

A The Filipino Scientist Fe del Mundo became known throughout the world because she invented medical devices for children. ✕ Your answer

B Fe del Mundo became known throughout the world for inventing medical devices for children. ✓ Correct answer

C The Filipino Scientist Fe del Mundo, who invented medical devices for children, became known throughout the world.

D The Filipino Scientist became known throughout the world after she invented medical devices for children.

Explanation

B keeps the named person and the main relationship between her invention work and worldwide recognition in a compact sentence. “For inventing” states the reason without an extra clause. A and C are wordier, while D removes her name. The answer evaluates the wording of the supplied statement, rather than independently establishing the historical claim.

Question 29 · Language Skills ✕ Wrong

The measure of choosing well is whether or not man likes what he _____.

A Choose ✕ Your answer

B Is choosing

C Chosen

D Chose ✓ Correct answer

Explanation

The source intends “chose,” the simple past tense of “choose”: a person evaluates whether they like a choice already made. “Choose” does not agree with “he,” and “chosen” needs an auxiliary verb. However, “is choosing” also forms a grammatical sentence about a current choice, so the stem needs a past-time cue before D can be uniquely correct.

Question 30 · Language Skills ✓ Correct

The breakthrough discovery happened out of the blue, shocking the entire scientific community. What does it mean?

A Expectedly

B From the ocean

C Completely unexpectedly ✓ Your answer

D With great effort

Explanation

“Out of the blue” means unexpectedly, with no advance indication. The clue that the discovery shocked the community reinforces that meaning. It does not refer to the ocean, an expected outcome, or how much effort the work required.

Question 31 · Language Skills ✓ Correct

Read the passage and answer the question based on the passage.

Passage C

Enzymes are specialized biological catalysts composed of protein chains. They accelerate biochemical reactions by lowering the activation energy needed for a reaction to occur. Each enzyme possesses an active site that fits perfectly with a specific molecule called a substrate, resembling a lock-and-key mechanism. Environmental factors like temperature and pH levels can alter an enzyme's shape, rendering it nonfunctional through a process called denaturation.

What happens to an enzyme during the process of "denaturation"?

A It multiplies rapidly.

B Its shape changes, making it nonfunctional. ✓ Your answer

C It transforms into a substrate.

D Its activation energy increases.

Explanation

In the passage, denaturation changes the enzyme's shape and leaves it nonfunctional. Because the active site depends on shape, the described change prevents normal functioning. The passage does not define denaturation as multiplication, transformation into a substrate, or an increase in activation energy.

Question 32 · Language Skills ✕ Wrong

Read the passage and answer the question based on the passage.

Passage C
Enzymes are specialized biological catalysts composed of protein chains. They accelerate biochemical reactions by lowering the activation energy needed for a reaction to occur. Each enzyme possesses an active site that fits perfectly with a specific molecule called a substrate, resembling a lock-and-key mechanism. Environmental factors like temperature and pH levels can alter an enzyme's shape, rendering it nonfunctional through a process called denaturation.

The term "active site" refers to the part of the enzyme that:

A Produces heat energy ✕ Your answer

B Combines with the substrate ✓ Correct answer

C Destroys neighboring proteins

D Measures cellular acidity

Explanation
The active site is described as the part that fits a specific substrate. Choice B best matches this binding or combining role. The passage does not describe the active site as producing heat, destroying nearby proteins, or measuring acidity.

Question 33 · Language Skills ✕ Wrong

Identify the figure of speech most clearly illustrated in the sentence.

“The wind howled through the forest, whistling an eerie tune through the trees.”

A Simile

B Personification ✓ Correct answer

C Metaphor ✕ Your answer

D Oxymoron

Explanation

“Whistling an eerie tune” presents the wind as if it were a person making music. Giving a nonhuman natural force such a human action is personification. There is no explicit “like” or “as” comparison, no direct identity comparison, and no paired contradiction to support the other choices.

Question 34 · Language Skills ✓ Correct

Choose the word closest in meaning to the capitalized word as used in the sentence.

Her speech was full of REDUNDANT phrases that made the presentation longer than necessary.

A Unique

B Repetitive ✓ Your answer

C Concise

D Informative

Explanation

“Redundant” means unnecessarily repeated or not needed because the same idea has already been expressed. Repetitive phrases explain why the speech became longer than necessary. “Unique” means distinct, “concise” means brief without wasted words, and “informative” means providing useful information.

Question 35 · Language Skills ✓ Correct

The city planted more trees along the highway. After a year, the air quality in the area improved. What is the most likely reason for the improvement?

A Cars stopped using the highway.

B Trees helped reduce air pollution. ✓ Your answer

C The weather became colder.

D People avoided traveling.

Explanation

The passage gives two linked observations: the city planted more trees, and air quality later improved. The most reasonable explanation among the choices is that the trees helped reduce air pollution. The passage does not say that traffic stopped, temperatures dropped, or people avoided travel. This is an inference from the information given, not proof that trees were the only cause.

Question 36 · Language Skills ✓ Correct

Identify the option containing an error in the following sentence or passage. If there is no error, choose D.

In order for a parasitic fungi to thrive and proliferate, they spread spores around themselves, hoping that an unfortunate victim passes by and catches some spores.

A In order for a parasitic fungi to thrive and proliferate, they spread spores around themselves, ✓ Your answer

B hoping that an unfortunate victim

C passes by and catches some spores.

D No error

Explanation

“A” requires a singular count noun, but “fungi” is plural. One consistent correction is “In order for parasitic fungi to thrive and proliferate, they spread spores around themselves,” removing “a.” This keeps “fungi,” “they,” and “themselves” plural. Alternatively, singular “a parasitic fungus” would require “it spreads” and “itself.” Both linked segments are in choice A.

Question 37 · Language Skills ✓ Correct

Read the passage and answer the question based on the passage.

Passage C

Enzymes are specialized biological catalysts composed of protein chains. They accelerate biochemical reactions by lowering the activation energy needed for a reaction to occur. Each enzyme possesses an active site that fits perfectly with a specific molecule called a substrate, resembling a lock-and-key mechanism. Environmental factors like temperature and pH levels can alter an enzyme's shape, rendering it nonfunctional through a process called denaturation.

Which of the following environmental factors is explicitly mentioned as a potential cause for denaturation?

A Atmospheric pressure

B pH levels ✓ Your answer

C Salinity

D Light intensity

Explanation

The passage names temperature and pH as environmental factors that can change enzyme shape. Of the choices, only pH levels appears in that statement. The question asks what is explicitly mentioned, so possible effects of other conditions are outside the evidence being tested.

Question 38 · Language Skills ✓ Correct

Read the passage and answer the question based on the passage.

Passage A

The concept of the "carrying capacity" is fundamental to ecological science. It refers to the maximum population size of a biological species that a specific environment can sustain indefinitely, given the food, water, habitat, and other available necessities. When a population exceeds its carrying capacity, resources become depleted, leading to a population crash or die-off until equilibrium is restored.

According to Passage A, what determines the carrying capacity of an environment?

A The physical size of the animals

B The available necessities like food and water ✓ Your answer

C The reproductive rate of the biological species

D The migration patterns of predators

Explanation

The passage defines carrying capacity by the resources an environment can provide, explicitly including food, water, habitat, and other necessities. Choice B restates this defining condition. The size of animals, their reproductive rate, and predator migration are not given as the definition in the passage.

Question 39 · Language Skills ✓ Correct

Securing funding for the engineering laboratory was no easy task; they had to pull strings behind the scenes. What does it mean?

A Use secret influence or connections ✓ Your answer

B Unravel a complex fabric pattern

C Play a musical string instrument

D Tie loose knots together

Explanation

“Pull strings” means use influence or personal connections to obtain a result, often through private arrangements. “Behind the scenes” reinforces that meaning. The other choices interpret strings literally as fabric, musical strings, or knots rather than recognizing the idiom.

Question 40 · Language Skills ✕ Wrong

Identify the option containing an error. If there is no error, choose D.

Neither the principal investigators nor the project director were available to sign the clearance form yesterday.

A were ✓ Correct answer

B clearance form ✕ Your answer

C yesterday

D No error

Explanation

With subjects joined by “neither...nor,” the verb normally agrees with the nearer subject. Here that subject is singular “project director,” so the correct verb is “was.” The earlier plural “principal investigators” does not determine agreement in this word order.

Question 41 · General Sciences ✓ Correct

Sound waves are an example of what type of wave?

A Transverse wave

B Longitudinal wave ✓ Your answer

C Electromagnetic wave

D Standing wave

Explanation

Sound waves travel through compressions and rarefactions, with particles vibrating parallel to the direction of wave travel - the defining feature of longitudinal waves.

Question 42 · General Sciences ✓ Correct

Two runners cover the same distance, but Runner A takes less time than Runner B. Which statement is correct?

A Runner A has lower average speed

B Runner A has higher average speed ✓ Your answer

C Both runners have the same average speed

D Average speed cannot be determined

Explanation

Since speed = distance/time, covering the same distance in less time means a higher average speed.

Question 43 · General Sciences ✓ Correct

Two forces of 8 N and 5 N act on an object in the same direction. What is the net force?

A 3 N

B 5 N

C 8 N

D 13 N ✓ Your answer

Explanation

Since both forces act in the same direction, they simply add: 8 N + 5 N = 13 N.

Question 44 · General Sciences ✕ Wrong

What type of tectonic stress causes rocks to slide horizontally past one another?

A Compression Stress

B Confining Stress

C Tension Stress ✕ Your answer

D Shear Stress ✓ Correct answer

Explanation

Shear stress occurs when rock masses are subjected to opposing horizontal forces that cause them to slide past one another. This type of stress is responsible for the formation of transform faults such as the San Andreas Fault in California.

Question 45 · General Sciences ✓ Correct

Which sequence correctly represents the major flow of genetic information in a typical cell?

A RNA → DNA → protein

B Protein → RNA → DNA

C DNA → RNA → protein ✓ Your answer

D DNA → protein → RNA

Explanation

In the central dogma, information generally flows from DNA to RNA through transcription and from RNA to protein through translation.

Question 46 · General Sciences ✕ Wrong

What is the pH of a 0.001 M HCl solution?

A 1 ✕ Your answer

B 2

C 3 ✓ Correct answer

D 4

Explanation

Hydrochloric acid (HCl) is a strong acid that dissociates completely in aqueous solutions. The pH is defined as -log[H+]. For a 0.001 M solution, [H+] = 1 × 10^-3 M, so pH = -log(1 × 10^-3) = 3. Reference: Chang, R., & Overby, J. (2019). Chemistry (13th ed.). McGraw-Hill Education.

Question 47 · General Sciences ✓ Correct

A person has difficulty breathing because the alveoli in the lungs have become damaged. Which function is most directly affected?

A Production of red blood cells

B Exchange of oxygen and carbon dioxide ✓ Your answer

C Digestion of proteins

D Filtration of blood

Explanation

Alveoli provide a large surface area for gas exchange between air and blood. Damage to them compromises oxygen uptake and carbon dioxide removal.

Question 48 · General Sciences — ✕ Wrong

An object starts from rest and accelerates uniformly at 2 m/s². What is its velocity after 5 seconds?

A 2.5 m/s — ✕ Your answer

B 5 m/s

C 10 m/s — ✓ Correct answer

D 20 m/s

Explanation

v = u + at = 0 + (2)(5) = 10 m/s.

Question 49 · General Sciences — ✕ Wrong

According to the theory of Seafloor Spreading, where on the ocean floor would you expect to find the youngest oceanic crust?

A Deep inside ocean trenches near subduction zones — ✕ Your answer

B Along the crest and axis of mid-ocean ridges — ✓ Correct answer

C On abyssal plains adjacent to continental margins

D Beneath thick stacks of accumulated marine sediments

Explanation

A new ocean floor is constantly being created right at the middle of the ocean, where underwater mountains called mid-ocean ridges sit. The newest rock stays closest to that ridge, while older rock slowly moves outward toward the continents.

Question 50 · General Sciences ✓ Correct

A circuit has a resistance of 5 ohms and a current of 2 amperes. What is the voltage?

A 2.5 V

B 7 V

C 10 V ✓ Your answer

D 25 V

Explanation

V = IR = (2)(5) = 10 V.

Question 51 · General Sciences ✓ Correct

Which formula correctly represents electric power?

A P=IR

B P=VI ✓ Your answer

C P=V/I

D P=I/V

Explanation

Electric power is the product of voltage and current: P=VI (derived from combining P=W/t and Ohm's Law).

Question 52 · General Sciences ✗ Wrong

Which condition would provide the strongest evidence that a geothermal field can produce electricity sustainably?

A High surface temperature caused by recent volcanic activity ✕ Your answer

B Abundant hot springs supplied by shallow groundwater

C A permeable hot reservoir with sufficient heat and continuous fluid recharge ✓ Correct answer

D Thick volcanic rocks capable of retaining heat near the surface

Explanation

Sustainable geothermal energy production requires three essential components: a consistent heat source, permeable rock formations that allow fluid circulation, and a reliable supply of water that is naturally replenished. Without continuous recharge, the system will gradually cool and cease operation.

Question 53 · General Sciences ✕ Wrong

A 10 kg object experiences a net force of 20 N. What is its acceleration?

A $0.5\ m/s^2$

B $2\ m/s^2$ ✓ Correct answer

C $10\ m/s^2$ ✕ Your answer

D $200\ m/s^2$

Explanation

$F = ma \rightarrow a = F/m = 20/10 = 2\ m/s^2$.

Question 54 · General Sciences ✕ Wrong

A patient has insufficient antidiuretic hormone (ADH). Which effect would most likely occur?

A Increased water reabsorption by the kidneys

B Decreased urine production ✕ Your answer

C Increased water loss through dilute urine ✓ Correct answer

D Complete cessation of filtration in the kidneys

Explanation

ADH promotes water reabsorption in the kidneys. Insufficient ADH therefore results in less water being reabsorbed and a greater volume of dilute urine.

Question 55 · General Sciences ✓ Correct

A researcher wants to determine whether a newly discovered organism is more closely related to Species X or Species Y. Which evidence would generally provide the strongest molecular comparison?

A Comparing their body sizes only

B Comparing their preferred habitats only

C Comparing homologous DNA or protein sequences ✓ Your answer

D Comparing their common names

Explanation

Homologous DNA or protein sequences provide molecular evidence that can be used to infer evolutionary relationships. Greater sequence similarity generally indicates closer evolutionary relationship, provided appropriate sequences are compared.

Question 56 · General Sciences ✕ Wrong

A person eats a carbohydrate-rich meal. Blood glucose rises, stimulating the pancreas to release insulin. Insulin promotes glucose uptake by cells, causing blood glucose to decrease. This is an example of:

A Positive feedback ✕ Your answer

B Negative feedback ✓ Correct answer

C Genetic recombination

D Osmoregulation through diffusion

Explanation

Negative feedback counteracts an initial change. Rising blood glucose stimulates insulin release, which lowers blood glucose toward its normal range.

Question 57 · General Sciences — ✕ Wrong

What is the molecular geometry of a methane (CH_4) molecule according to VSEPR theory?

A Linear — ✕ Your answer

B Trigonal planar

C Tetrahedral — ✓ Correct answer

D Bent

Explanation

According to VSEPR theory, methane (CH4) has four single bonds and zero lone pairs around the central carbon atom. These four electron domains repel each other equally, resulting in a tetrahedral geometry with bond angles of 109.5°. Reference: Chang, R., & Overby, J. (2019). Chemistry (13th ed.). McGraw-Hill Education.

Question 58 · General Sciences — ✓ Correct

A geologist examines a mineral that is abundant in many rocks and is known for being relatively resistant to chemical weathering. Over long periods, less-resistant minerals break down while this mineral remains and accumulates in riverbeds, beaches, and deserts, becoming a major component of ordinary sand. Which mineral is being described?

A Gypsum

B Calcite

C Mica

D Quartz — ✓ Your answer

Explanation

Quartz is one tough mineral that barely reacts with water or chemicals. While other minerals break down and wash away, quartz stays right where it is. Over time, it piles up in riverbeds, beaches, and deserts, and that is why most sand is made of quartz.

Question 59 · General Sciences — ✕ Wrong

What is the molality (m) of a solution containing 1.0 mole of solute dissolved in 500 grams of solvent?

A 0.5 m

B 1.0 m

C 2.0 m — ✓ Correct answer

D 5.0 m — ✕ Your answer

Explanation

Molality (m) = moles of solute / mass of solvent (in kg). Convert solvent mass: 500 g = 0.5 kg. Then m = 1.0 mol / 0.5 kg = 2.0 m. Reference: LibreTexts. (2016). 16.11: Molality.

Question 60 · General Sciences — ✓ Correct

A researcher observes a cell containing a nucleus, mitochondria, endoplasmic reticulum, and Golgi apparatus. Which conclusion is most appropriate?

A The cell is prokaryotic.

B The cell is eukaryotic. — ✓ Your answer

C The cell cannot perform cellular respiration.

D The cell lacks membrane-bound organelles.

Explanation

The presence of a nucleus and membrane-bound organelles such as mitochondria, ER, and Golgi apparatus identifies the cell as eukaryotic.

Question 61 · General Sciences — ✓ Correct

A person suddenly enters a very cold environment. Which response best illustrates homeostasis?

A The body permanently lowers its internal temperature.

B The body increases heat production and reduces heat loss. ✓ Your answer

C The body stops cellular respiration.

D The body prevents blood from circulating to all tissues.

Explanation

Maintaining a relatively stable internal temperature despite external changes is homeostasis. Increasing heat production and reducing heat loss help restore internal temperature.

Question 62 · General Sciences ✕ Wrong

A scientist measures mountains from their base on the seafloor rather than from sea level. Which mountain would have a different ranking compared with measurements based on elevation above sea level?

A Mauna Loa ✓ Correct answer

B Mount Everest ✕ Your answer

C Mount Kilimanjaro

D Mount Elbrus

Explanation

Most mountains are measured from sea level, but Mauna Loa starts way down at the bottom of the Pacific Ocean. If you measure it from its true base underwater, it is actually taller than Mount Everest even though its peak does not stand as high above the ocean surface.

Question 63 · General Sciences ✓ Correct

A scientist measures enzyme activity at different temperatures. Activity increases from 10°C to 37°C but decreases sharply at 60°C. Which explanation is most likely?

A The enzyme becomes permanently more active at all temperatures.

B High temperature may disrupt the enzyme's three-dimensional structure. ✓ Your answer

C Enzymes cannot function below 60°C.

D Temperature changes the DNA sequence coding for the enzyme immediately.

Explanation

Enzymes depend on specific three-dimensional structures. Excessive heat can disrupt bonds maintaining that structure, reducing catalytic activity or denaturing the enzyme.

Question 64 · General Sciences ✓ Correct

Which set of quantum numbers represents an electron in a 3p orbital?

A $n=3, l=0, m_l=0$

B $n=3, l=1, m_l=-1$ ✓ Your answer

C $n=3, l=2, m_l=+1$

D $n=2, l=1, m_l=0$

Explanation

For a 3p orbital, the principal quantum number is n=3, and the angular momentum quantum number for a p orbital is l=1. The magnetic quantum number m_l can range from -l to +l, meaning -1, 0, or +1. Therefore, n=3, l=1, $m_l=-1$ is a valid set. Reference: Chang, R., & Overby, J. (2019). Chemistry (13th ed.). McGraw-Hill Education.

Question 65 · General Sciences ✗ Wrong

Rapid urbanization in CALABARZON has led to the paving over of prime agricultural lands with concrete and asphalt. This drastically alters soil hydrology. What is this called?

A soil salinization

B soil sealing ✓ Correct answer

C soil compaction ✗ Your answer

D laterization

Explanation

When we pave over farmland and natural ground with concrete and asphalt, we are sealing the soil. Water cannot soak in anymore, and the soil loses its ability to breathe, store moisture, or support plant life.

Question 66 · General Sciences ✓ Correct

Earth's natural greenhouse effect maintains planetary habitability. Which physical mechanism describes how greenhouse gases (e.g., CO_2 CH_4 H_2O vapor) warm the lower atmosphere?

A They absorb outgoing longwave thermal infrared radiation emitted by Earth's surface and re-emit heat in all directions. ✓ Your answer

B They reflect 100% of incoming solar ultraviolet light back into space.

C They generate heat energy spontaneously through nuclear fusion in the troposphere.

D They trap high-frequency cosmic rays inside polar ice caps.

Explanation

Greenhouse gases absorb outgoing longwave thermal infrared radiation emitted by Earth's surface and re-radiate heat downward.

Question 67 · General Sciences ✕ Wrong

In pea plants, tallness (T) is dominant over dwarfness (t). If two heterozygous plants are crossed, what proportion of their offspring is expected to be dwarf?

A 0%

B 25% ✓ Correct answer

C 50% ✕ Your answer

D 75%

Explanation

The cross is Tt×Tt. The expected genotypes are TT, Tt, Tt, and tt. Thus, 1 out of 4, or 25%, is expected to be a dwarf.

Question 68 · General Sciences ✓ Correct

In the reaction 2K + Cl2 ⟶ 2KCl, which element is reduced?

A K

B Cl2 ✓ Your answer

C K+

D Cl-

Explanation

In the reaction 2K + Cl2 ⟶ 2KCl, chlorine goes from an oxidation state of 0 to -1, gaining electrons and therefore being reduced. (C) and (D) are incorrect because they do not exist on the reactant side.

Question 69 · General Sciences ✕ Wrong

Which gas law states that volume is directly proportional to temperature?

A Boyle's Law ✕ Your answer

B Charles' Law ✓ Correct answer

C Avogadro's Law

D Dalton's Law

Explanation

According to Charles's Law, volume is directly proportional to absolute temperature when pressure is held constant. Boyle's Law is a pressure-volume relationship. Avogadro's Law is a volume-mole relationship. Dalton's Law deals with partial pressures.

Question 70 · General Sciences ✓ Correct

What is the half-life (t½) of a first-order nuclear decay process with a rate constant k = 0.693 day^{-1}?

A 1.0 day ✓ Your answer

B 2.0 days

C 3.0 days

D 4.0 days

Explanation

For a first-order process, half-life is given by t½ = ln(2) / k ≈ 0.693 / k. With k = 0.693 day^-1, t½ = 0.693 / 0.693 day^-1 = 1.0 day. Reference: Chang, R., & Overby, J. (2019). Chemistry (13th ed.).

Question 71 · General Sciences ✓ Correct

Which structure in a plant is primarily responsible for absorbing water and mineral ions from the soil?

A Flower

B Root hair ✓ Your answer

C Stoma

D Xylem vessel

Explanation

Root hairs greatly increase the surface area of roots, facilitating absorption of water and mineral ions from the soil.

Question 72 · General Sciences ✓ Correct

The Philippines is designated as a global biodiversity hotspot due to its high concentration of endemic species (organisms found nowhere else on Earth). Which ecological factor is primarily responsible for the evolution of high endemism in archipelagic ecosystems?

A High frequency of volcanic disturbances preventing plant succession

B Geographical isolation of islands inhibiting gene flow and driving speciation over evolutionary time ✓ Your answer

C Homogeneous climate conditions across islands eliminating specialized ecological niches

D Decreased mutation rates among island populations compared to mainland species

Explanation

Oceanic barriers isolate island populations from mainland gene pools. This long-term reproductive and geographic isolation, combined with unique local selective pressures and adaptive radiation, leads to the evolution of distinct species adapted to specific island environments.

Question 73 · General Sciences ✕ Wrong

Why are microplastics (plastic fragments smaller than 5mm) considered particularly harmful environmental contaminants?

A They biodegrade completely within hours, releasing volatile methane gas. ✕ Your answer

B Their high surface area and hydrophobic surfaces adsorb toxic persistent organic pollutants (POPs) from seawater. ✓ Correct answer

C They lower soil electrical conductivity, preventing root ion uptake.

D They dissolve readily in rainwater to form strong mineral acids.

Explanation

Microplastics persist long-term and adsorb hydrophobic persistent organic pollutants (POPs) from surrounding water.

Question 74 · General Sciences ✓ Correct

A plant is placed in darkness for several days. Which process would be most directly reduced?

A Photosynthesis ✓ Your answer

B Glycolysis

C DNA replication

D Protein digestion

Explanation

Photosynthesis requires light energy during its light-dependent reactions. Extended darkness therefore directly prevents or greatly reduces photosynthetic activity.

Question 75 · General Sciences ✓ Correct

In the History of Earth, which Era is called the "Age of Dinosaurs"?

A Paleozoic

B Mesozoic ✓ Your answer

C Cenozoic

D Precambrian

Explanation

The Mesozoic Era, spanning approximately 252 million to 66 million years ago, was the period during which dinosaurs emerged, diversified, and dominated terrestrial ecosystems. It is therefore widely referred to as the Age of Dinosaurs.

Question 76 · General Sciences ✗ Wrong

Which bond results from the transfer of electrons?

A Covalent ✗ Your answer

B Metallic

C Ionic ✓ Correct answer

D Hydrogen

Explanation

Ionic bonding occurs through the complete electrostatic transfer of one or more valence electrons from a metal (low electronegativity) to a nonmetal (high electronegativity), forming

oppositely charged cations and anions. Reference: Zumdahl, S. S., & Zumdahl, S. A. (2018). Chemistry (10th ed.). Cengage Learning. (Chapter 8: Chemical Bonds).

Question 77 · General Sciences ✓ Correct

Republic Act 6969 regulates toxic substances and hazardous and nuclear wastes in the Philippines. What requirement does RA 6969 place on chemical importers and hazardous waste generators?

A Priority Chemical List (PCL) compliance, chemical pre-manufacture notifications, and hazardous waste manifest tracking. ✓ Your answer

B Exemption from environmental impact assessments inside economic zones.

C Mandatory disposal of nuclear waste into shallow coastal marine trenches.

D Replacement of safety data sheets (SDS) with LGU business permits

Explanation

RA 6969 mandates chemical pre-notifications, Priority Chemical Lists, and hazardous waste tracking manifests.

Question 78 · General Sciences ✕ Wrong

Which metamorphic rock texture forms specifically under differential stress, resulting in the parallel alignment of platy minerals such as mica?

A Aphanitic texture ✕ Your answer

B Foliated texture ✓ Correct answer

C Granoblastic texture

D Porphyroblastic texture

Explanation

When rocks are subjected to directed pressure and elevated temperatures during regional metamorphism, platy minerals such as mica align in a parallel orientation. This produces a distinct layered or banded appearance known as foliation.

Question 79 · General Sciences ✕ Wrong

Which organic compound is an isomer of ethanol (CH_3CH_2OH)?

A Methanol (CH_3OH) ✕ Your answer

B Acetone (CH_3COCH_3)

C Acetic acid (CH_3COOH)

D Dimethyl ether (CH_3OCH_3) ✓ Correct answer

Explanation

Ethanol (CH3CH2OH) and dimethyl ether (CH3OCH3) have the same molecular formula (C2H6O) but different structural connectivity of their atoms, making them constitutional (structural) isomers. Reference: LibreTexts. (2021). 2.4: Isomerization in Organic Compounds.

Question 80 · General Sciences ✓ Correct

How many significant figures are in 0.004060 m?

A 2

B 3

C 4 ✓ Your answer

D 5

Explanation

Leading zeros (the zeros before the 4) are placeholders and are never significant. Captive zeros between non-zero numbers (the zero between 4 and 6) are always significant. Trailing zeros to the right of the decimal point (the final zero) are also significant. Therefore, the significant figures are 4, 0, 6, and 0 (4 significant figures).

Question 81 · Mathematics ✓ Correct

Find the slope of a line parallel to $5y - 2x = 7$.

A $m = \frac{2}{5}$ ✓ Your answer

B $m = -\frac{2}{5}$

C $m = \frac{5}{2}$

D $m = -\frac{5}{2}$

Explanation

$$5y - 2x = 7$$

becomes

$$5y = 2x + 7,$$

so

$$y = \frac{2}{5}x + \frac{7}{5}.$$

Therefore,

$$\boxed{m = \frac{2}{5}}.$$

Question 82 · Mathematics — Not answered

Three radar stations A, B, and C form a triangle. $AB = 7\text{ km}$, $AC = 8\text{ km}$, and $\angle BAC = 60^\circ$. What is the exact distance BC?

A 57 km

B $\sqrt{57}\text{ km}$ ✓ Correct answer

C 13 km

D $\sqrt{113}\text{ km}$

Explanation

$BC^2 = 7^2 + 8^2 - 2(7)(8)\cos 60^\circ = 57 \implies BC = \sqrt{57}\,\text{km}.$

Question 83 · Mathematics — Not answered

What is the measured angle of $\angle A$?

A 165°

B 48°

C 156° ✓ Correct answer

D 39°

Explanation

Identify tangent lines: Lines CB and CD are tangent to the circle centered at A from external point C. By geometric properties, a tangent line is perpendicular to the radius drawn to the point of tangency. Therefore:

$\angle ABC = 90^\circ$

$\angle ADC = 90^\circ$

Use quadrilateral interior angles: In quadrilateral ABCD, the interior angles sum to 360°:

$\angle BAD + \angle ABC + \angle BCD + \angle ADC = 360^\circ$

Substitute known values:

$$\angle BAD + 90^\circ + 24^\circ + 90^\circ = 360^\circ$$
$$\angle BAD + 204^\circ = 360^\circ$$
$$\angle BAD = 156^\circ$$

Question 84 · Mathematics — Not answered

Find the inverse of the function $f(x) = \sqrt[3]{x+1} + 2$.

A $f^{-1}(x) = (x-2)^3 - 1$ ✓ Correct answer

B $f^{-1}(x) = (x+2)^3 - 1$

C $f^{-1}(x) = (x-2)^3 + 1$

D $f^{-1}(x) = \sqrt[3]{x-1} + 2$

Explanation

To find the inverse of $f(x) = \sqrt[3]{x+1} + 2$, replace $f(x)$ with y:

$$y = \sqrt[3]{x+1} + 2.$$

Swap x and y and solve for y:

$$x = \sqrt[3]{y+1} + 2.$$

$$x - 2 = \sqrt[3]{y+1}.$$

Cube both sides:

$$(x-2)^3 = y + 1.$$

$$y = (x-2)^3 - 1.$$

Therefore,

$$\boxed{f^{-1}(x) = (x-2)^3 - 1}.$$

Question 85 · Mathematics — Not answered

Triangle ABC has side lengths $a = 10$, $b = 14$, and included angle $C = 150^\circ$. What is the exact area of triangle ABC?

A 35 ✓ Correct answer

B $35\sqrt{3}$

C 70

D $70\sqrt{3}$

Explanation

Area $= \frac{1}{2}(10)(14)\sin(150^\circ) = 35.$

Question 86 · Mathematics — Not answered

A sector of a circle has a radius of 12 cm and a central angle of 60°. Find the length of the arc bounding the sector.

A 2π cm

B 4π cm ✓ Correct answer

C 6π cm

D 8π cm

Explanation

The formula for the length(s) given a central angle (in degrees) is:
$s = \frac{\theta}{360^\circ} \times 2\pi r$
Substitute the given values $(r = 12, \theta = 60^\circ)$:

$$s = \frac{60^\circ}{360^\circ} \times 2\pi(12)$$
$$s = \frac{1}{6} \times 24\pi = 4\pi \text{ cm}$$

Question 87 · Mathematics — Not answered

A chord 16 cm long is balanced inside a circle exactly 6 cm away from the center. What is the total area of the circle?

A 64π cm^2

B 80π cm^2

C 100π cm^2 ✓ Correct answer

D 121π cm^2

Explanation

A perpendicular line drawn from the center of a circle to a chord bisects the chord.
- Half length of the chord $= \frac{16}{2} = 8$ cm
- Distance from center = 6 cm

These two measurements form the legs of a right triangle where the hyphotenuse is the circle's radius(r):

$$r^2 = 6^2 + 8^2$$
$$r^2 = 36 + 64 = 100$$
$$r = 10 \text{ cm}$$

Computing the area:
$Area = \pi r^2 = \pi(10)^2 = 100\pi \text{ cm}^2$

Question 88 · Mathematics — Not answered

What is the exact value of $\tan(75^\circ) + \tan(15^\circ)$?

A $2 - \sqrt{3}$

B 2

C $2\sqrt{3}$

D 4 ✓ Correct answer

Explanation
$\tan(75^\circ) + \tan(15^\circ) = (2 + \sqrt{3}) + (2 - \sqrt{3}) = 4.$

Question 89 · Mathematics — Not answered

Consider the rational function $f(x) = \dfrac{x^2 - 9}{x - 3}$. Which statement about its graph is TRUE?

A The graph has a vertical asymptote at $x = 3$.

B The graph has a hole at $x = 3$. ✓ Correct answer

C The graph has an x-intercept at $x = 3$.

D The graph has no discontinuities.

Explanation

Factor:

$$x^2 - 9 = (x - 3)(x + 3).$$

Thus,

$$f(x) = \frac{(x - 3)(x + 3)}{x - 3} = x + 3.$$

However, $x = 3$ remains excluded because it makes the original denominator zero. Therefore, the graph has a hole at

$$\boxed{x = 3}.$$

Question 90 · Mathematics — Not answered

What is the exact value of $\cos(105^\circ)$?

A $\frac{\sqrt{2}-\sqrt{6}}{4}$ ✓ Correct answer

B $\frac{\sqrt{6}-\sqrt{2}}{4}$

C $\frac{\sqrt{2}+\sqrt{6}}{4}$

D $\frac{-\sqrt{2}-\sqrt{6}}{4}$

Explanation

$\cos(45^\circ + 60^\circ) = \frac{\sqrt{2}}{2} \cdot \frac{1}{2} - \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{2}-\sqrt{6}}{4}$.

Question 91 · Mathematics — Not answered

The quadratic function $f(x) = x^2 + kx + 9$ is restricted to the domain $x \geq a$ so that its inverse is also a function. For which value of a should the domain begin?

A $a = -\frac{k}{2}$ ✓ Correct answer

B $a = \frac{k}{2}$

C $a = k$

D $a = -k$

Explanation

For

$$f(x) = x^2 + kx + 9,$$

the vertex occurs at

$$x = -\frac{b}{2a}.$$

Since $a = 1$ and $b = k$,

$$x = -\frac{k}{2}.$$

Thus, the restricted domain should begin at

$$\boxed{a = -\frac{k}{2}}.$$

Question 92 · Mathematics — Not answered

What is the principal value of $(1 - i\sqrt{3})^6$ expressed in rectangular form?

A $-64i$

B 64 ✓ Correct answer

C -64

D $64i$

Explanation

$1 - i\sqrt{3} = 2e^{-i\pi/3}$. Then $\left(2e^{-i\pi/3}\right)^6 = 64e^{-2\pi i} = 64.$

Question 93 · Mathematics — Not answered

$P(x) = 2x + kx^2 - 4x + 5$ is divided by $(x - 2)$, the remainder is 13. Find the value of k.

A -2

B -3

C 3 ✓ Correct answer

D 2

Explanation

Using the Remainder Theorem,

$$P(2) = 13.$$

If

$$P(x) = kx^2 - 2x + 5,$$

then

$$P(2) = 4k - 4 + 5.$$

Thus,

$$4k + 1 = 13,$$

$$4k = 12,$$

and

$$\boxed{k = 3}.$$

Question 94 · Mathematics — Not answered

The total area of a regular hexagon is $24\sqrt{3}\ \text{cm}^2$. What is the length of one of its sides?

A 2 cm

B 4 cm ✓ Correct answer

C 6 cm

D 8 cm

Explanation

A regular hexagon consists of 6 equilateral triangles. The area formula is:
$Area = \frac{3\sqrt{3}}{2}s^2$
Equate it to the given area to find side length(s):
$24\sqrt{3} = \frac{3\sqrt{3}}{2}s^2$
Divide both sides by $\sqrt{3}$:

$$24 = \frac{3}{2}s^2$$
$$48 = 3s^2$$
$$s^2 = 16 \rightarrow s = 4 \text{ cm}$$

Question 95 · Mathematics — Not answered

Using power-reduction formulas, express $\sin^4 x$ in terms of first powers of cosines.

A $\frac{3+4\cos(2x)+\cos(4x)}{8}$

B $\frac{3-4\cos(2x)+\cos(4x)}{8}$ ✓ Correct answer

C $\frac{1-2\cos(2x)+\cos(4x)}{4}$

D $\frac{3-2\cos(2x)+\cos(4x)}{8}$

Explanation

$\sin^4 x = \left(\frac{1-\cos 2x}{2}\right)^2 = \frac{1-2\cos 2x+\cos^2 2x}{4} = \frac{3-4\cos 2x+\cos 4x}{8}$.

Question 96 · Mathematics — Not answered

Line AB and CD intersect at point E as displayed below. Which among the pair choices are supplementary angles?

A $\angle CEB$ and $\angle AED$

B $\angle AEC$ and $\angle CEB$

C $\angle CEB$ and $\angle BED$

D B and C ✓ Correct answer

Explanation

Supplementary Angles: Two or more angles on a straight line form a straight angle, which equals 180° (a linear pair or supplementary sum).

$$\angle CEB \text{ and } \angle AED \rightarrow \text{Vertical Angles}$$
$$\angle AEC \text{ and } \angle CEB \rightarrow \text{Supplementary Angles}\checkmark$$
$$\angle CEB \text{ and } \angle BED \rightarrow \text{Supplementary Angles}\checkmark$$

Question 97 · Mathematics — Not answered

A fashion designer bought a lot of suits for ₱750. He sold all but 3 of them for ₱864 making a profit of ₱7 on each suit sold. How many suits did he buy?

A 20 suits

B 22 suits

C 24 suits

D 30 suits ✓ Correct answer

Explanation

Let x be the number of suits bought. Each cost $\frac{750}{x}$ pesos, and the $x-3$ suits sold went for $\frac{864}{x-3}$ pesos each. A profit of ₱7 on each suit sold gives

$$\frac{864}{x-3} - \frac{750}{x} = 7.$$

Multiplying by $x(x-3)$:

$$864x - 750(x-3) = 7x(x-3)$$
$$7x^2 - 135x - 2250 = 0$$
$$(x-30)(7x+75) = 0$$

So $x = 30$: he bought 30 suits at ₱25 each and sold 27 of them at ₱32 each.

Question 98 · Mathematics — Not answered

Consider the function $f(x) = \frac{1}{2\sin x - 1}$ on the domain $[0, 2\pi]$. Which statement correctly describes the behavior of $f(x)$ near its vertical asymptotes?

A $f(x) \to +\infty$ from both sides of $x = \frac{\pi}{6}$.

B $f(x) \to -\infty$ from both sides of $x = \frac{5\pi}{6}$.

C $f(x) \to +\infty$ as $x \to \left(\frac{\pi}{6}\right)^-$ and $f(x) \to -\infty$ as $x \to \left(\frac{\pi}{6}\right)^+$.

D $f(x) \to -\infty$ as $x \to \left(\frac{\pi}{6}\right)^-$ and $f(x) \to -\infty$ as $x \to \left(\frac{5\pi}{6}\right)^+$. ✓ Correct answer

Explanation

Asymptotes occur where $\sin x = \frac{1}{2}$, i.e., $x = \frac{\pi}{6}, \frac{5\pi}{6}$. For x just below $\frac{\pi}{6}$ and just above $\frac{5\pi}{6}$, $2\sin x - 1 < 0$, so $f(x) \to -\infty$.

Question 99 · Mathematics — Not answered

Evaluate the exact value of $\cos\left(\arcsin\left(-\frac{3}{5}\right) + \arccos\left(-\frac{5}{13}\right)\right)$.

A $-\frac{56}{65}$

B $-\frac{16}{65}$

C $\frac{16}{65}$ ✓ Correct answer

D $\frac{56}{65}$

Explanation

Let $\alpha = \arcsin\left(-\frac{3}{5}\right)$ and $\beta = \arccos\left(-\frac{5}{13}\right)$, so $\cos\alpha = \frac{4}{5}$ and $\sin\beta = \frac{12}{13}$. Then $\cos(\alpha + \beta) = \left(\frac{4}{5}\right)\left(-\frac{5}{13}\right) - \left(-\frac{3}{5}\right)\left(\frac{12}{13}\right) = \frac{16}{65}$.

Question 100 · Mathematics — Not answered

Simplify $(8i)(-2i)(-2-8i)$.

A $-32-128i$ ✓ Correct answer

B $32-128i$

C $32+128i$

D $16-64i$

Explanation

$$(8i)(-2i)(-2-8i).$$

First,

$$(8i)(-2i) = -16i^2.$$

Since

$$i^2 = -1,$$

we get

$$16.$$

Then,

$$16(-2-8i) = -32-128i.$$

Therefore,

$$\boxed{-32-128i}.$$

Question 101 · Mathematics — Not answered

Which term refers to the coefficient of the highest-degree term in a polynomial?

A Constant term

B Leading coefficient ✓ Correct answer

C Degree

D Monomial

Explanation

The leading coefficient is the coefficient attached to the term with the highest degree.

$$\boxed{\text{Leading coefficient}}.$$

Question 102 · Mathematics — Not answered

Which of the following is the slope-intercept form of the equation of the line through the point $(5, -1)$ with the slope $= -\frac{3}{5}$?

A $y = -\frac{3}{5}x + 2$ ✓ Correct answer

B $y = \frac{3}{5}x + 2$

C $y = -\frac{3}{5}x - 2$

D $y = -\frac{5}{3}x + 2$

Explanation

Using point-slope form,

$$y - y_1 = m(x - x_1).$$

Substitute the given point and slope, then simplify into slope-intercept form.

Question 103 · Mathematics — Not answered

A cistern can be filled by one pipe in 20 minutes and by another in 30 minutes. How long will it take both pipes together to fill the tank?

A 10 minutes

B 12 minutes ✓ Correct answer

C 15 minutes

D 25 minutes

Explanation

The first pipe fills

$$\frac{1}{20}$$

of the tank per minute, while the second fills

$$\frac{1}{30}.$$

Together:

$$\frac{1}{20}+\frac{1}{30}=\frac{3}{60}+\frac{2}{60}=\frac{1}{12}.$$

Therefore,

$$\boxed{12 \text{ minutes}}.$$

Question 104 · Mathematics — Not answered

The surface area S of a cylinder is $S=2\pi r^2+2\pi rh$ where r is the base radius and h is the height. What is h, in inches, when $S=175$ square inches and $r=6$ inches?

A $\frac{175-864\pi^2}{12\pi}$

B $\frac{175+72\pi}{12\pi}$

C $\frac{175+864\pi^2}{12\pi}$

D $\frac{175-72\pi}{12\pi}$ ✓ Correct answer

Explanation

Using

$$S = 2\pi r^2 + 2\pi rh,$$

substitute $S = 175$ and $r = 6$:

$$175 = 2\pi(6)^2 + 2\pi(6)h.$$

Thus,

$$175 = 72\pi + 12\pi h.$$

Solving for h gives

$$h = \frac{175 - 72\pi}{12\pi} \approx 1.51.$$

Therefore,

$$\boxed{h \approx 1.51 \text{ inches}}.$$

Question 105 · Mathematics — Not answered

Reduce the following radical: $\sqrt[10]{64x^8y^4}$.

A $2^{3/5}x^{4/5}y^{2/5}$ ✓ Correct answer

B $2^3x^4y^2$

C $2^{1/3}x^{2/3}y^2$

D $2^3xy^{4\bullet 2}$

Explanation

$$64 = 2^6.$$

Thus,

$$\sqrt[10]{64x^8y^4} = (2^6x^8y^4)^{1/10}.$$

Therefore,

$$2^{6/10}x^{8/10}y^{4/10} = \boxed{2^{3/5}x^{4/5}y^{2/5}}.$$

Question 106 · Mathematics — Not answered

Evaluating the equation $\sin(x+y)\sin(x-y) = \sin^2 x - \sin^2 y$, which statement correctly describes this relation?

A. It is a valid trigonometric identity true for all real values of x and y. ✓ Correct answer

B. It is a conditional equation true only when $x = y$.

C. It is an identity true only when $x, y \in [0, \pi/2]$.

D. It is a conditional equation true only when $x + y = \frac{\pi}{2}$.

Explanation

Expanding both sides yields $\sin^2 x - \sin^2 y$. It is a valid identity for all real x and y.

Question 107 · Mathematics — Not answered

How many sides have a polygon if the sum of the interior angles is $1080°$?

A. 5

B. 6

C. 7

D. 8 ✓ Correct answer

Explanation

The sum of the interior angles of a polygon is given by the formula:

$$S = (n-2) \times 180°$$

Given that the sum of the interior angles is 1080°:

$$(n-2) \times 180^\circ = 1080^\circ$$

Divide both sides by 180°:

$$n - 2 = 6$$

Add 2 to both sides:

$$n = 8$$

Therefore, the polygon has 8 sides.

Question 108 · Mathematics — Not answered

Loki has an 18-meter ladder leaning against a vertical wall. The distance of the foot of the ladder from the wall is one-fourth of the ladder's length. How high up the wall does the ladder reach?

A $\frac{9\sqrt{15}}{2}$ m ✓ Correct answer

B $\frac{9\sqrt{17}}{2}$ m

C $\frac{9\sqrt{15}}{4}$ m

D $\frac{9\sqrt{17}}{4}$ m

Explanation

Use the Pythagorean Theorem ($c^2 = a^2 + b^2$), where the ladder is $c = 18$ m and its foot is $a = \frac{1}{4}(18) = \frac{9}{2}$ m from the wall:

$$18^2 = \left(\frac{9}{2}\right)^2 + b^2$$
$$b^2 = 324 - \frac{81}{4} = \frac{1215}{4}$$
$$b = \frac{\sqrt{1215}}{2} = \frac{9\sqrt{15}}{2} \text{ m}$$

Question 109 · Mathematics — Not answered

An angle $\theta = \frac{17\pi}{3}$ is drawn in standard position. Which quadrant does its terminal side lie in, and what is its reference angle θ'?

A Quadrant I, $\theta' = \frac{\pi}{3}$

B Quadrant II, $\theta' = \frac{2\pi}{3}$

C Quadrant IV, $\theta' = \frac{\pi}{3}$ ✓ Correct answer

D Quadrant IV, $\theta' = \frac{\pi}{6}$

Explanation

$\frac{17\pi}{3} = 6\pi - \frac{\pi}{3} \equiv -\frac{\pi}{3}$, which lies in Quadrant IV. Reference angle $\theta' = \frac{\pi}{3}$.

Question 110 · Mathematics — Not answered

The distance between $(8, -7)$ and $(x, 5)$ is 13. Find x.

A 2

B 3 ✓ Correct answer

C 4

D 5

Explanation

Using the distance formula:
$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Where:

$$(x_2, y_2) = (x, 5)$$
$$(x_1, y_1) = (8, -7)$$

Substitute:

$$\begin{aligned} 13 &= \sqrt{(x-8)^2+(5-(-7))^2} \\ (13)^2 &= (\sqrt{(x-8)^2+144})^2 \\ 13^2 &= (x-8)^2+144 \\ 169-144 &= (x-8)^2 \\ \sqrt{25} &= \sqrt{(x-8)^2} \\ \pm 5 &= x-8 \\ x &= 3, x = 13 \end{aligned}$$

Observing the choices above, it is clear that the answer is 3.

Question 111 · Mathematics — Not answered

What is the exact value of $\tan\left(2\arcsin\left(\frac{1}{3}\right)\right)$?

A $\frac{\sqrt{2}}{3}$

B $\frac{2\sqrt{2}}{7}$

C $\frac{4\sqrt{2}}{7}$ ✓ Correct answer

D $\frac{3\sqrt{2}}{4}$

Explanation

Let $\alpha = \arcsin\left(\frac{1}{3}\right)$, so $\tan\alpha = \frac{\sqrt{2}}{4}$. Then $\tan(2\alpha) = \frac{2\left(\frac{\sqrt{2}}{4}\right)}{1-\frac{1}{8}} = \frac{4\sqrt{2}}{7}$.

Question 112 · Mathematics — Not answered

What is the number of real solutions to $2\sin^2 x + 3\cos x - 3 = 0$ on the interval $[0, 2\pi]$?

A 1

B 2

C 3

D 4 ✓ Correct answer

Explanation

Using $\sin^2 x = 1 - \cos^2 x$: $(2\cos x - 1)(\cos x - 1) = 0$. $\cos x = 1 \implies x = 0, 2\pi$; $\cos x = \frac{1}{2} \implies x = \frac{\pi}{3}, \frac{5\pi}{3}$. Total: 4.

Question 113 · Mathematics — Not answered

What is the exact value of $\sec^2\left(\frac{11\pi}{6}\right) - \tan^2\left(\frac{7\pi}{4}\right) + \sin\left(\frac{3\pi}{2}\right)$?

A $-\frac{2}{3}$ ✓ Correct answer

B $\frac{1}{3}$

C $-\frac{1}{3}$

D $\frac{2}{3}$

Explanation

$\left(\frac{2}{\sqrt{3}}\right)^2 - (-1)^2 + (-1) = \frac{4}{3} - 1 - 1 = -\frac{2}{3}$.

Question 114 · Mathematics — Not answered

For which value of k does the function $f(x) = \dfrac{(k-3)x + 2}{x + 1}$ have no inverse function?

A $k = -2$

B $k = 3$

C $k = 4$

D $k = 5$ ✓ Correct answer

Explanation

If $k = 5$, then

$$f(x) = \frac{2x + 2}{x + 1} = 2 \quad \text{for every } x \neq -1.$$

A constant function is not one-to-one, so it has no inverse. For $k = -2$, 3 or 4, the numerator is not a multiple of $x + 1$, so f is one-to-one and has an inverse.

Question 115 · Mathematics — Not answered

When solving $\sqrt{3}\tan x = 2\sin x$ on $[0, 2\pi)$, a student divides both sides by $\sin x$ to obtain $\frac{\sqrt{3}}{\cos x} = 2 \implies \cos x = \frac{\sqrt{3}}{2}$, yielding $x = \frac{\pi}{6}, \frac{11\pi}{6}$. Which evaluation of the student's solution is correct?

A. The student lost two valid solutions ($x = 0$ and $x = \pi$) by dividing by $\sin x$ without considering when $\sin x = 0$. ✓ Correct answer

B. The student's method is completely correct and all solutions were found.

C. The student introduced 2 extraneous solutions by dividing by $\sin x$.

D. The student made an error in evaluating $\cos x = \frac{\sqrt{3}}{2}$.

Explanation

Dividing by $\sin x$ assumes $\sin x \neq 0$, losing the solutions $x = 0$ and $x = \pi$.

Question 116 · Mathematics — Not answered

Given $\sin\theta = -\frac{4}{5}$ with θ in Quadrant III, what is the exact value of $\tan(2\theta)$?

A. $\frac{24}{7}$

B. $\frac{7}{24}$

C. $-\frac{24}{7}$ ✓ Correct answer

D. $-\frac{7}{24}$

Explanation

In Quadrant III, $\cos\theta = -\frac{3}{5}$, so $\tan\theta = \frac{4}{3}$. Then $\tan(2\theta) = \frac{2\left(\frac{4}{3}\right)}{1-\frac{16}{9}} = -\frac{24}{7}$.

Question 117 · Mathematics — Not answered

What is the maximum value of $f(x) = 5\sin x - 12\cos x + 4$?

A 13

B 15

C 17 ✓ Correct answer

D 21

Explanation

The amplitude of $5\sin x - 12\cos x$ is $\sqrt{5^2 + 12^2} = 13$. Maximum $= 13 + 4 = 17$.

Question 118 · Mathematics — Not answered

The volume of a cube is $216\ \text{cm}^3$. Find the total surface area of the cube.

A $144\ \text{cm}^2$

B $180\ \text{cm}^2$

C $216\ \text{cm}^2$ ✓ Correct answer

D $256\ \text{cm}^2$

Explanation

First, find the edge length (s) of the cube using the volume formula:

$V = s^3 \rightarrow 216\ \text{cm}^3 = s^3 \rightarrow s = 6\ \text{cm}$

Now calculate the total surface area (TSA), which is composed of 6 square faces:

$$TSA = 6s^2$$
$$= 6(6)^2 = 6(36) = 216\ \text{cm}^2$$

Question 119 · Mathematics — Not answered

A sector of a circle of radius r has a perimeter equal to half the circumference of the entire circle. What is the area of this sector in terms of r?

A $\left(\frac{\pi-1}{2}\right) r^2$

B $\left(\frac{\pi-2}{2}\right) r^2$ ✓ Correct answer

C $(\pi - 2)r^2$

D $\left(\frac{2\pi-1}{4}\right) r^2$

Explanation

Perimeter: $2r + r\theta = \pi r \implies \theta = \pi - 2$. Area: $A = \frac{1}{2}r^2\theta = \left(\frac{\pi-2}{2}\right) r^2$.

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$$3x + 2y - 7 = 0 \quad \text{eqn. 1}$$
$$2x - By + 2 = 0 \quad \text{eqn. 2}$$

Rearranging eqn. 1 to slope-intercept form...
$y = -\frac{3}{2}x + \frac{7}{2}; \quad m = -\frac{3}{2} \Rightarrow -m^{-1} = \frac{2}{3}$

Rearranging eqn. 2 to slope-intercept form...
$y = \frac{2}{B}x + \frac{2}{B}$

By inspection, relating the slope of eqn. 2 to $(-m^{-1})$ of eqn. 1, $B = 3$.

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